SciOdyssey.

Autonomous ML Research Agent

Preserve the research world. Reset the researcher.

01 / PROBLEM · FAILURE MODES

Why current research agents fall short.

Three obstacles to sustained autonomous research.

01 CLOSED-WORLD WORKFLOW

Observe | Hypothesize | Code | Evaluate

× ERROR

dead end

Predefined tools and rigid paths cannot handle the unexpected.

Unexpected problems require developer intervention.

02 SINGLE TRAJECTORY

same basin

A long-lived context carries yesterday's momentum into today's search.

More iterations ≠ more exploration.

03 TREE SEARCH

UNDERSTAND → EDIT → READ CONTEXT →

repeat

Reconstructing file context costs far more tokens than the edit itself.

Context cost forces repeated reading.

DESIGN RESPONSE — Research is open-world, non-linear, and exploratory. ADAPT · DIVERSIFY · LEARN

02 / MOTIVATION

Research world

PERSISTENT / ON DISK

EVIDENCE — **Knowledge + experiment ledger**
Metrics, failures and original reports

CURATED — **Research brief + working priors**
Working hypotheses; open to revision

CODE — **Retained implementation + State**
Recoverable version with evidence and provenance

↓ CONTEXT RESET — Inherit experience, not prior reasoning.

03 / SYSTEM

One world. Two scientific roles.

ENVIRONMENT
Task contract
Curated data + evaluation
Starter baseline
Budget + permissions

→

RESEARCH WORLD
Verified evidence
Knowledge ledger
Research brief
Implementation + State

CONTEXT RESET
inherit experience, not prior reasoning

FRESH SCIENTIST
reads world
forms hypotheses
runs experiments
reports evidence

→

META REVIEW
audits claims
scopes conclusions
curates memory
preserves optional State

↻ Selective persistence / trajectory handoff
Audit → scope → preserve → fresh Scientist

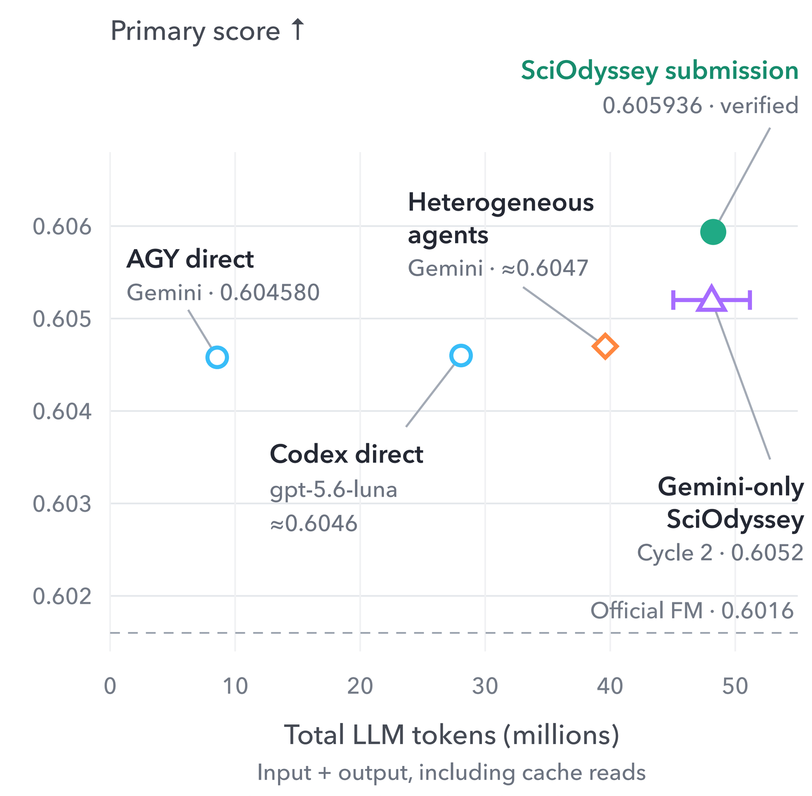
Runtime / deterministic mechanics
Workspace isolation · runner + evaluator · logging · artifact capture · State operations

PARALLEL BREADTH isolated branches → post-hoc review → explicit adoption

05 / EVALUATION

Research outcomes, in context.

KuaiRand-Pure · public validation · one recorded run per condition



* Comparative exploration across distinct agent setups under KuaiRand-Pure benchmark constraints.

06 / VERIFIED SUBMISSION · PUBLIC VALIDATION

Verified submission · public validation

Metric	Official FM	SciOdyssey	Absolute Δ
Primary	**0.6016000**	**0.6059363**	+0.0043363
GAUC	0.6674000	0.6728421	+0.0054421
nDCG@5	0.5357000	0.5390304	+0.0033304

Primary = mean(GAUC, nDCG@5). Final recipe: 46 categorical features × eight-seed FM ensemble · NumPy / CPU.

4 autonomous cycles · **13** named experiments · **0** manual scientific interventions*

7 Full evaluations · 0 GPU-hours · 170,588 prediction rows passed the unchanged Starter Kit alignment checker.

* 100% autonomous execution throughout the retained submission record.

07 / INSIGHTS

Observed behavior, not just a score.

Recovery

After evaluation, NumPy float32 values broke serialization. The Scientist repaired the writer; META retained the evidence for the next trajectory.

Audit

Reviews caught leakage and metric issues early. META audits empirical claims against artifacts before promoting State.

Insight synthesis. Heterogeneous Scientists turn diverse search styles into a broader exploration prior; parallel search produces stronger candidates.

04 / EXPERIENCE

Start your journey.

Autonomous by default. Supervisable by design.

`task.md`
What to solve
Objectives · constraints · evaluation

`PERSONAL.md`
How you want it solved
Preferences · research style · priorities

Define the task → Delegate research → Review evidence

Coding agent — EXAMPLE INPUT

```
› Use the research-agent skill.
  Follow task.md and PERSONAL.md.
  Let me review the results and evidence.
```

github.com/zc6600/research-agent-kuairand ↗

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Sources: slides/slides.md · docs/ARCHITECTURE.md · docs/FINAL_REPORT.md §§3–5 · Code, report, ledger and checked output in repository.